

Request:

22279	51.4323899...	172.64.149.23	192.168.31.206	TCP	1005 [TCP Previous segment not captured] 80 → 6001
22282	51.4324810...	172.64.149.23	192.168.31.206	HTTP	1005 [TCP Retransmission] HTTP/1.1 200 OK (applic
41070	278.568127...	192.168.31.206	34.223.124.45	HTTP	515 GET / HTTP/1.1
41073	278.853597...	34.223.124.45	192.168.31.206	TCP	887 [TCP Previous segment not captured] 80 → 6138

An HTTP request is being sent to 34.223.124.45 IP destination.

Status is 515 GET/HTTP/1.1

This highlights the request for information

For this request, there are 5 header layers

```
> Frame 41070: Packet, 515 bytes on wire (4120 bits), 515 bytes captured (4120 bits) on inte
> Ethernet II, Src: CyberTANTech_a6:ae:03 (00:45:e2:a6:ae:03), Dst: XiaomiMobile_7d:01:57 (8
> Internet Protocol Version 4, Src: 192.168.31.206, Dst: 34.223.124.45
> Transmission Control Protocol, Src Port: 61389, Dst Port: 80, Seq: 1, Ack: 1, Len: 461
> Hypertext Transfer Protocol
```

1. Frame 41070:

```
Frame 41070: Packet, 515 bytes on wire (4120 bits), 515 bytes captured (4120 bits) on inte
  Section number: 1
  > Interface id: 0 (\Device\NPF_{80FA1A5B-845E-4127-A53C-4ECAD5ADAAFF})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 10, 2026 10:50:50.297735000 Bangladesh Standard Time
  UTC Arrival Time: Mar 10, 2026 04:50:50.297735000 UTC
  Epoch Arrival Time: 1773118250.297735000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 278.200 microseconds]
  [Time delta from previous displayed frame: 3 minutes, 47.135646900 seconds]
  [Time since reference or first frame: 4 minutes, 38.568127900 seconds]
  Frame Number: 41070
  Frame Length: 515 bytes (4120 bits)
  Capture Length: 515 bytes (4120 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:tcp:http]
  Character encoding: ASCII (0)
  [Coloring Rule Name: HTTP]
  [Coloring Rule String: http || tcp.port == 80 || http2]
```

The total size of the frame is 515 bytes on the wire. And data is encapsulated.

The actual data is segmented into frames

2. Ethernet II

```
Ethernet II, Src: CyberTANTech_a6:ae:03 (00:45:e2:a6:ae:03), Dst: XiaomiMobile_7d:01:57 (8c:de:f9:7d:01:57)
> Destination: XiaomiMobile_7d:01:57 (8c:de:f9:7d:01:57)
> Source: CyberTANTech_a6:ae:03 (00:45:e2:a6:ae:03)
Type: IPv4 (0x0800)
[Stream index: 0]
```

This layer helps to send frames on my local network. As the destination and source MAC addresses are visible.

3. IPv4:

```
Internet Protocol Version 4, Src: 192.168.31.206, Dst: 34.223.124.45
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 501
  Identification: 0xecad (60589)
> 010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0x8cd2 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.31.206
  Destination Address: 34.223.124.45
  [Stream index: 119]
```

This layer handles end-to-end routing of the packets using Source IP and Destination IP addresses.

4. TCP:

```
Transmission Control Protocol, Src Port: 61389, Dst Port: 80, Seq: 1, Ack: 1, Len: 461
  Source Port: 61389
  Destination Port: 80
  [Stream index: 177]
  [Stream Packet Number: 6]
> [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 461]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 2221669134
  [Next Sequence Number: 462 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 3646336873
  0101 .... = Header Length: 20 bytes (5)
> Flags: 0x018 (PSH, ACK)
  Window: 255
  [Calculated window size: 65280]
  [Window size scaling factor: 256]
  Checksum: 0xfd0a [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
> [Timestamps]
> [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (461 bytes)
```

The TCP layer ensures the connection-oriented transmission of packets. Source port, which has to be random from the client end. And the destination port for the web has to be 80. The window size 255 indicates how much data it can hold.

5. HTTP:

```
Hypertext Transfer Protocol
> GET / HTTP/1.1\r\n
Host: neverssl.com\r\n
Connection: keep-alive\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/svg+xml
Referer: https://www.google.com/\r\n
Accept-Encoding: gzip, deflate\r\n
Accept-Language: en-US,en;q=0.9\r\n
\r\n
[Full request URI: http://neverssl.com/]
```

This layer shows the hostname of the website and the type of request it sends. This request shows the user asking for the root webpage.

Response:

41300	282.397603...	192.168.31.206	34.223.124.45	HTTP	546 GET /online HTTP/1.1
41307	282.678695...	34.223.124.45	192.168.31.206	HTTP	603 [HTTP/1.1 301 Moved Permanently] (text/html)
41311	282.682400...	192.168.31.206	34.223.124.45	HTTP	547 GET /online/ HTTP/1.1
41323	282.962900...	34.223.124.45	192.168.31.206	TCP	133 [TCP Previous segment not captured] 80 → 65212 [PSH, ACK]

This is the response status from the website. The response is coming with text or HTML files.

As requested, we will have 5 layers and one extra layer for line-based text data

```

> Frame 41307: Packet, 603 bytes on wire (4824 bits), 603 bytes captured (4824 bits) on interface 0
> Ethernet II, Src: XiaomiMobile_7d:01:57 (8c:de:f9:7d:01:57), Dst: CyberTANTech_a6:ae:03 (08:00:27:08:00:27)
> Internet Protocol Version 4, Src: 34.223.124.45, Dst: 192.168.31.206
> Transmission Control Protocol, Src Port: 80, Dst Port: 65212, Seq: 1, Ack: 493, Len: 549
> Hypertext Transfer Protocol
> Line-based text data: text/html (7 lines)

```

1. Frame:

```

v Frame 41307: Packet, 603 bytes on wire (4824 bits), 603 bytes captured (4824 bits) on interface 0
  Section number: 1
  > Interface id: 0 (\Device\NPF_{80FA1A5B-845E-4127-A53C-4ECAD5ADAAFF})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 10, 2026 10:50:54.408302200 Bangladesh Standard Time
  UTC Arrival Time: Mar 10, 2026 04:50:54.408302200 UTC
  Epoch Arrival Time: 1773118254.408302200
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 1.388300 milliseconds]
  [Time delta from previous displayed frame: 281.091300 milliseconds]
  [Time since reference or first frame: 4 minutes, 42.678695100 seconds]
  Frame Number: 41307
  Frame Length: 603 bytes (4824 bits)
  Capture Length: 603 bytes (4824 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:tcp:http:data-text-lines]
  Character encoding: ASCII (0)
  [Coloring Rule Name: HTTP]
  [Coloring Rule String: http || tcp.port == 80 || http2]

```

The total size of the frame received over the network is 603 bytes, with the encapsulation.

2. Ethernet II:

```

Ethernet II, Src: XiaomiMobile_7d:01:57 (8c:de:f9:7d:01:57), Dst: CyberTANTech_a6:ae:03 (08:00:27:08:00:27)
  > Destination: CyberTANTech_a6:ae:03 (00:45:e2:a6:ae:03)
  > Source: XiaomiMobile_7d:01:57 (8c:de:f9:7d:01:57)
  Type: IPv4 (0x0800)
  [Stream index: 0]

```

The second layer shows the local network delivery throughout the source MAC and destination MAC.

3. IPv4:

```

v Internet Protocol Version 4, Src: 34.223.124.45, Dst: 192.168.31.206
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 589
    Identification: 0x9e75 (40565)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 244
    Protocol: TCP (6)
    Header Checksum: 0x66b2 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 34.223.124.45
    Destination Address: 192.168.31.206
    [Stream index: 119]

```

The IPv4 layer shows logical routing across the internet. These addresses reflect the inbound direction.

4. TCP:

```

v Transmission Control Protocol, Src Port: 80, Dst Port: 65212, Seq: 1, Ack: 493, Len: 549
  Source Port: 80
  Destination Port: 65212
  [Stream index: 187]
  [Stream Packet Number: 8]
  > [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 549]
    Sequence Number: 1 (relative sequence number)
    Sequence Number (raw): 2065653146
    [Next Sequence Number: 550 (relative sequence number)]
    Acknowledgment Number: 493 (relative ack number)
    Acknowledgment number (raw): 1228871625
    0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
    Window: 219
    [Calculated window size: 28032]
    [Window size scaling factor: 128]
    Checksum: 0x1e7e [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  > [Timestamps]
  > [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (549 bytes)

```

In this layer, application-to-application delivery and reliability are ensured. As it is sent from the server, for that reason, the source port is 80, and the destination port is random, but here that is 65212.

5. HTTP:

```
✓ Hypertext Transfer Protocol
  > HTTP/1.1 301 Moved Permanently\r\n
    Date: Tue, 10 Mar 2026 04:50:52 GMT\r\n
    Server: Apache/2.4.62 ()\r\n
    Location: http://wonderoussoothingquietismile.neverssl.com/online/\r\n
  > Content-Length: 263\r\n
    Keep-Alive: timeout=5, max=100\r\n
    Connection: Keep-Alive\r\n
    Content-Type: text/html; charset=iso-8859-1\r\n
    \r\n
    [Request in frame: 41300]
    [Time since request: 281.091300 milliseconds]
    [Request URI: /online]
    [Full request URI: http://wonderoussoothingquietismile.neverssl.com/online]
    File Data: 263 bytes
```

This layer shows the actual server response and loads the content. But for this code, we see an **HTTP/1.1 301 Moved Permanently** status code. This means the server indicates that the page the client wants that go to new address.

6. Line-based textdata:

```
✓ Line-based text data: text/html (7 lines)
  <!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">\n
  <html><head>\n
  <title>301 Moved Permanently</title>\n
  </head><body>\n
  <h1>Moved Permanently</h1>\n
  <p>The document has moved <a href="http://wonderoussoothingquietismile.neverssl.com/onli
  </body></html>\n
```

This is the actual body message of the HTTP response. The message body comes immediately after the HTTP headers. This is the 7-line raw HTML code that helps to load the website